Multiple Choice:

1. Evaluate $\lim_{x\to\infty}\frac{3x^2+2x+1}{x^2+x+1}$.

   A. 2
   B. 3
   C. 4
   D. 1

2. Find the equation of the tangent line to $y = x^3 - 2x^2 + 2$ at x = 1.

   A. $y = -x + 2$
   B. $y = x - 2$
   C. $y = x + 2$
   D. $y = -x - 2$

3. Water is flowing into a conical vessel 15 cm. deep and having a radius of 3.75 cm. across the top. If the rate at which water is rising is 2 cm/sec. How fast is the water flowing into the conical vessel when the depth of water is 4 cm?

   A. $2.8\ m^3/min$
   B. $6.28\ m^3/min$
   C. $8.26\ m^3/min$
   D. $6.38\ m^3/min$

4. Find the minimum amount of tin sheet that can be made into a closed cylinder having a volume of 108 cu. inches in square inches.

   A. 125.50
   B. 127.50
   C. 129.50
   D. 123.50

5. Given a cone of diameter x and altitude of h. What percent is the volume of the largest cylinder which can be inscribed in the cone to the volume of the cone?

   A. 44 %
   B. 46 %
   C. 56 %
   D. 65 %

6. A balloon is released from the ground 100 meters from an observer. The balloon rises directly upward at the rate of 4 meters per second. How fast is the balloon receding from the observer 10 seconds later?

   A. 1.68 m/sec
   B. 1.36 m/sec
   C. 1.55 m/sec
   D. 1.49 m/sec

7. A box is to be constructed from a piece of zinc 20 sq.in by cutting equal squares from each corner and turning up the zinc to form the side. What is the volume of the largest box that can be so constructed?

   A. 599.95 cu in.
   B. 592.59 cu in.
   C. 579.50 cu in.
   D. 622.49 cu in.

8. A statue 3 m high is standing on a base of 4 m high. If an observer's eye is 1.5 m above the ground, how far should he stand from the base in order that the angle subtended by the statue is a maximum.

   A. 3.41 m
   B. 3.51 m
   C. 3.71 m
   D. 4.41 m

9. Water is running into a hemispherical bowl having a radius of 10 cm at a constant rate of 3 cm^3/min. When the water is x cm. deep, the water level is rising at the rate of 0.0149 cm/min. What is the value of x?

   A. 3
   B. 2
   C. 4
   D. 5

10. Evaluate: $\lim_{x\to1}\frac{x^2-1}{x^2+3x-4}$

    A. 1/5
    B. 2/5
    C. 3/5
    D. 4/5

11. Evaluate: $\lim_{x\to0}\frac{1-\cos x}{x^2}$

    A. 0
    B. 1/2
    C. 2
    D. –1/2

12. What is the derivative with respect to x of $(x + 1)^3 - x^3$?

    A. 3x + 6
    B. 3x – 3
    C. 6x – 3
    D. 6x + 3

13. Find y' if y = arcsin cos x

    A. – 1
    B. – 2
    C. 1
    D. 2

14. Differentiate the equation: $y = \frac{x^2}{x+1}$

    A. $\frac{x^2+2x}{(x+1)^2}$
    B. $\frac{x}{x+1}$
    C. 2x
    D. $\frac{2x^2}{x+1}$

15. Give the slope of the curve at the point (1,1): $y = \frac{x^3}{4} - 2x + 1$

    A. 1/4
    B. –1/4
    C. 1 1/4
    D. –1 ¼

16. Find the equation of the normal to $x^2 + y^2 = 5$ at the point (2,1).

    A. y = 2x
    B. x = 2y
    C. 2x + 3y = 3
    D. x + y = 1

17. At point of inflection,

    A. y' = 0
    B. y" = 0
    C. y" is negative

D. y” is positive

18. The _______ derivative of the function is the rate of change of the slope of the graph.

A. first
B. second
C. third
D. fourth

19. Two posts, one 8 m and the other 12 m high are 15 cm apart. If the posts are supported by a cable running from the top of the first posts to a stake on the ground and then to the top of the second post, find the distance from the lower post to the stake to use minimum amount of cable.

A. 6 m from the lower post
B. 5 m from the lower post
C. 4 m from the lower post
D. 3 m from the lower post

20. The distance a body travels is a function of time t and is defined by: $x(t) = 18t + 9t^2$. What is its velocity at t = 3?

A. 70
B. 72
C. 74
D. 76

21. When the area in square units of an expanding circle is increasing twice as fast as its radius in linear unit, the radius is ____.

A. π
B. $1/\pi$
C. 2π
D. $1/2\pi$

22. The depth of water in a cylindrical tank 4 m in diameter is increasing at the rate of 0.7 m/min. Find the rate at which the water flowing into the tank.

A. 7.3291 cu. m/min
B. 8.8302 cu. m/min
C. 8.3201 cu. m/min
D. 8.7964 cu. m/min

23. Evaluate $\frac{d}{dx}\left(\ln e^{2x}\right)$.

A. 1
B. – 2
C. – 1
D. 2

24. An open rectangular box with square base is to be made from 48 sq. m of material. What is the largest possible volume of the box in cubic meters?

A. 28
B. 30
C. 32
D. 34

25. The volume of a cylindrical tin can with a top and a bottom is to be 16π cubic inches. If a minimum amount of tin is to be used to construct the can, what must be the height, in inches, of the can?

A. 4
B. 5
C. 6
D. 8

END